

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

We are given :

- Cost to plant the tree : \$ 90 (one-time cost)
- Each year , the tree produces 7 lemons
- Each lemon sells for \$ 1 . 5 0
- Annual cost to water and feed : \$ 3

We want to find out **after how many years** Carlos starts **earning money** , that is , when the income from selling lemons exceeds the costs of planting and maintaining the tree .

Let's define :

- Let y be the number of years after which Carlos starts earning money .
- Annual income from lemons : $7 \times 1.5 = 10.5$ \$ dollars
- Annual expenses : \$ 3
- So , net income per year after the initial planting cost : $10.5 - 3 = 7.5$ \$ dollars per year

But since the initial \$ 90 cost is incurred upfront , we need to find when the **total earnings** (net of expenses) over y years **covers** the initial \$ 90 cost .

So , we set up the equation :

$$\text{Total net income over } y \text{ years} = 7.5 y$$

We want this to equal or exceed the initial cost :

$$7.5 y \geq 90$$

Solve for y :

$$y \geq \frac{90}{7.5} = 12$$

So , after **12 years** , the total earnings cover the initial cost . Therefore , **after the 12 th year** , Carlos starts earning money (since in the 13 th year , he is making a profit beyond the cost). But the question is : *When does he start earning money ?*

This means the **first year in which he makes a profit** , i.e. , when cumulative earnings exceed the initial cost .

After 12 years , the total net profit is exactly zero (he breaks even). So in year 13 , he starts making a *profit* (since revenue exceeds expenses , and the cost has been covered).

But the question says : *"How many years will it take before he starts earning money on the lemon tree ?"*

It might be interpreted as : after how many years of operation does he begin to make **profit** ? So , **at year 13** ?

Wait , but actually , if we interpret this question as when he **starts earning money** (i.e. , profits begin to happen) , then this is when the **net income first turns positive** .

So at year 1 , he incurs \$ 90 upfront , then makes \$ 7 . 5 per year after .

So **after year 12** , the net cumulative income is $7.5 \times 12 = 90$ \$, which **covers** the initial cost . At that point , the **profit is zero** , he has not yet started *earning* money (he hasn't made a profit , just broken even).

So he **starts earning money** only after that , i.e. , **after the 12 th year** , so in the **13 th year** .

Wait , but some interpretations of "start earning money" could mean when the cumulative income exceeds the planting cost . So **after 12 years** , so it takes 12 years to break even .

Therefore , he **starts earning money** after the initial cost is recovered .

Thus , **after 12 years** , he no longer loses money , at year 13 , he earns profit .

So when does he **start earning money** ?

Answer : after **12 years** , when the cost is recovered .

So , he starts earning money after **12 years** , meaning **it takes 12 years** to break even .

So it takes 12 years to start earning money .

Let's check :

- After 1 year : $7.5 \times 1 = 7$